

Activity Guide — middle / module-07-eleven-multiplication

IKS Curriculum

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Activity 1 — Mental Math Sprint Cards

Module: $\times 11$ Multiplication — Ekādhikena Pūrveṇa · **Band:** Middle · **Duration:** 20 min

Purpose

Reinforce the day-by-day concepts of $\times 11$ Multiplication — Ekādhikena Pūrveṇa through a hands-on activity. Aligned with NEP 2020 §4.23 (experiential learning).

Materials

flashcards (20–50 pre-printed $\times 11$ problems), stopwatch

What students do

In pairs: one student holds up a card, the other answers within 3 seconds using the $\times 11$ method. Swap. Track personal best time.

Step-by-step

1. **Setup (5 min)** — gather materials; explain the task.
2. **Working phase (10 min)** — students work individually or in pairs.
3. **Share (5 min)** — selected students share their results with the class.

Differentiation

- **Extension:** ask advanced students to do an additional iteration or analysis.
- **Support:** pair students who need scaffolds; offer partial templates.

Assessment

This activity is **formative**. The teacher observes participation, accuracy, and reflection quality. No grade is assigned.

Safety

- General safety briefing before any materials are handled.

- If herbs, plants, or food are involved: kitchen-amount only, no ingestion of unknown preparations.
- Outdoor activities: stay in the designated area; supervised by the teacher.

Activity 01 — Mental $\times 11$ Speed Race

Module: $\times 11$ (*Ekādhikena Pūrveṇa*) · **Band:** Middle · **Day:** 5 · **Time:** 20 min

Type

Paired + group quick-fire drill

Goal

Each student does at least 20 mental $\times 11$ problems (mixed difficulty) in under 4 minutes, with peer-verified accuracy.

Materials per pair

- 1 printed flashcard deck of 24 problems (cut into separate cards if possible)
- 1 stopwatch
- Pen + paper for tally

Set-up (teacher, before class)

- Print and pre-cut the 24-problem flashcard deck. Mix the difficulty (8 no-carry, 8 carry, 8 three-digit).

Instructions (student-facing handout)

Speed Race — work in pairs across 4 rounds.

Round 1 (5 min, solo): Take 20 cards. Solve each silently in under 5 seconds. Self-check three random ones with long multiplication.

Round 2 (5 min, paired): Partner-A reads card 1 aloud. Partner-B answers in under 5 seconds. Swap roles. Continue until you’ve each done 10 problems out loud.

Round 3 (5 min, group chorus): Teacher displays a problem on the board. Class shouts the answer together. 10 problems total.

Round 4 (5 min, quick-fire): Teacher calls a 2-digit number. First student to call back the $\times 11$ product gets a tally mark. Light competition — rotate the “first responder” so no single student dominates.

Flashcard deck (24 problems)

Easy (no-carry, 2-digit, 8 cards):

12×11	23×11	34×11	52×11
61×11	16×11	27×11	35×11

Medium (carry, 2-digit, 8 cards):

$$\begin{array}{cccc} 47 \times 11 & 58 \times 11 & 69 \times 11 & 76 \times 11 \\ 83 \times 11 & 89 \times 11 & 97 \times 11 & 99 \times 11 \end{array}$$

Hard (3-digit, 8 cards):

$$\begin{array}{cccc} 123 \times 11 & 234 \times 11 & 345 \times 11 & 456 \times 11 \\ 142 \times 11 & 365 \times 11 & 478 \times 11 & 529 \times 11 \end{array}$$

Answer key (teacher copy)

Easy: 132, 253, 374, 572, 671, 176, 297, 385

Medium: 517, 638, 759, 836, 913, 979, 1067, 1089

Hard: 1353, 2574, 3795, 5016, 1562, 4015, 5258, 5819

Discussion prompts (last 2 min)

- Which type was hardest? (Most students say “carry” or “3-digit.”)
- Did anyone discover a faster sub-trick?
- Are you noticeably faster than Day 1? (Yes — name a number.)

Looks-like / sounds-like

	Looks like	Sounds like
Going well	Pair alternating, one card per ~5 sec	“47 × 11!” — “517!” — “58 × 11!” — “638!”
Needs intervention	One partner stuck > 30 sec	Long silences; partner doing the math TO them, not WITH them

Safety / sensitivity

- Round 4 quick-fire risks dominating the room. If one student gets >4 tally marks while others have 0, switch to “must wait if you’ve answered the last 2.”
- No public leaderboards; tallies are private to each student.

Differentiation

- **Tier up:** Hard deck only. Try 4-digit × 11 (sample: 1234, 2345, 4567).
- **Tier down:** Easy deck only. Build no-carry fluency before adding carries.

Clean-up

- Worksheets kept by students. Flashcards return to the bag for re-use next year.

Activity 02 — *Anurūpyeṇa* Drill (×12 to ×19)

Module: ×11 (*Ekādhikena Pūrveṇa*) · **Band:** Middle · **Day:** 6 · **Time:** 20 min

Type

Solo + paired drill, mixed multipliers

Goal

Students mentally multiply 2-digit numbers by 12, 13, 14, 17, 19 using the *Anurūpyeṇa* pattern, with at least 80% accuracy.

Materials per pair

- 1 printed drill sheet (16 problems below)
- 1 stopwatch
- Pen each

Instructions (student-facing handout)

***Anurūpyeṇa* Drill** — work in pairs.

Round 1 (5 min, solo): Solve all 16 problems. No writing intermediate steps unless absolutely needed. Aim for the answer only.

Round 2 (5 min, paired): Partner-A reads problem 1. Partner-B answers within 10 seconds. Swap. Continue alternating through all 16.

Round 3 (5 min, mixed-multiplier): Teacher displays 8 new problems on the board, mixing all multipliers from $\times 11$ to $\times 19$. Class answers in chorus.

Round 4 (5 min, the hardest): 3 problems with $\times 19$ — the worst case. Show your working on paper this time.

Drill sheet (16 problems)

$\times 12$ problems:

1. 23×12 5. 41×12 9. 56×12 13. 78×12

$\times 13$ problems:

2. 24×13 6. 37×13 10. 45×13 14. 62×13

$\times 14$ problems:

3. 21×14 7. 35×14 11. 47×14 15. 53×14

$\times 17 + \times 19$ (harder):

4. 16×17 8. 23×17 12. 27×19 16. 31×19

Answer key (teacher copy)

$\times 12$: 276, 492, 672, 936

$\times 13$: 312, 481, 585, 806

$\times 14$: 294, 490, 658, 742

$\times 17 / \times 19$: 272, 391, 513, 589

Worked sanity check (give to slow pairs)

56×12 :

units = $6 \cdot 2 = 12 \rightarrow 2$, carry 1

tens = $5 \cdot 2 + 6 + 1 = 17 \rightarrow 7$, carry 1

hundreds = $5 + 1 = 6$

$\rightarrow 672 \checkmark$ (check: $56 \times 12 = 56 \times 10 + 56 \times 2 = 560 + 112 = 672$)

23×17 :

units = $3 \cdot 7 = 21 \rightarrow 1$, carry 2

tens = $2 \cdot 7 + 3 + 2 = 19 \rightarrow 9$, carry 1

hundreds = $2 + 1 = 3$

$\rightarrow 391 \checkmark$

Discussion prompts (last 2 min)

- Which multiplier was easiest? (Most students say $\times 12$.) Why?
- Which was the hardest? (Most say $\times 19$.) Why? (Bigger k means bigger carries.)
- How is the $\times 11$ case the “easy” version of this same pattern?

Looks-like / sounds-like

	Looks like	Sounds like
Going well	Pair calmly working through carries on paper	“Units 12, carry 1; tens 17 plus 6 plus 1 is 24, carry 2...”
Needs intervention	Pair guessing answers; no working shown	“I don’t know, maybe 600?” — “Same.”

Safety / sensitivity

- Same as Activity 01 — no public leaderboards.
- Some students will hit a wall at $\times 17/\times 19$. That’s expected — these are genuinely harder. Accept partial completion.

Differentiation

- **Tier up:** Try 3-digit $\times 13$ problems (e.g. 234×13). Same pattern, more digits.
- **Tier down:** Stick to $\times 12$ today. Build that fluency before adding $\times 13$, $\times 14$.

Clean-up

- Worksheets collected (teacher spot-checks before Day 7).

Project Brief

Project Brief — Module 7 Eleven Multiplication (Middle)

Work sessions: Day 9 · **Presentation:** Day 10 · **Total marks:** 35 (project 25 + presentation 10)

The project

Each pair (or individual, with teacher approval) chooses ONE of three formats:

Format 1: 60-second teaching video

Make a short video that teaches the $\times 11$ trick to a Grade 4 student. Constraints: - Total length: 60 seconds (± 10 sec). - Must show: at least 3 worked examples — one no-carry 2-digit, one carry case, one 3-digit. - Closing slide must cite Tirthaji (1965) AND a SutraGanita PDF.

Format 2: Printed workbook

Produce a 25-problem workbook with the $\times 11$ trick (plus 5 *Anurūpyeṇa* problems), designed for a Grade 4–5 self-learner. Constraints: - 25 problems organised in 4 difficulty levels: 7 no-carry 2-digit, 7 carry 2-digit, 6 three-digit, 5 *Anurūpyeṇa* ($\times 12$ – $\times 19$). - Each level includes ONE fully worked example. - Answer key on the back page. - Title page with cited source.

Format 3: Proof poster

Produce a one-page (A3) “proof poster” deriving $(10a + b) \times 11 = 100a + 10(a + b) + b$. Constraints: - The full derivation (at least 4 lines). - Three worked examples illustrating: no-carry, carry, cascading carry. - A 3–4 sentence “in plain English” explanation in your own words. - Bottom-strip citation of Tirthaji (1965) and the SutraGanita sūtra list PDF.

Why this matters

You’re moving from *learner* to *teacher* (or *publisher*, or *prover*). Each role forces you to understand more deeply.

The proof poster is the format that requires you to *own* the algebra. We recommend it if you struggled on Day 3.

What success looks like

A strong project will: - Demonstrate the $\times 11$ trick (or its scaled cousins) accurately, no arithmetic errors. - Show the algebraic identity OR derive it in detail. - Be pitched for the intended audience. - Cite the sources properly. - Show your own thinking, not just rephrase classroom examples.

Working in pairs

- Pair work strongly encouraged.
- Individual allowed for the workbook or poster format.
- Groups of 3 acceptable — each member must have a distinct role.

Citation rules

- At minimum **one** citation must appear in your project — visible.
- Format: Tirthaji, *Vedic Mathematics* (1965), sūtra 1 OR <PDF filename>, <page or section> (e.g. *Microsoft Word - sutras1.pdf*).

- See `sources.md` for the list of available references.

Self-reflection (submitted with project on Day 10)

In 4–6 sentences: 1. What did you set out to learn or make, and what did you actually end up with? 2. What was the most surprising thing you discovered while making this? 3. Did your view of “Vedic mathematics” change during this module? How?